

The day the meters learned to calibrate themselves

One bench session, six firmware releases: an eight-month-old fleet billing bug found and killed at its source, and a new self-mapping calibration architecture designed, built, broken, fixed, and certified against reference water.

200 gal → 0.13 gal

Phantom water per incident, before vs. after the stuck-event defenses — then eliminated entirely by the root-cause fix.

2 min 38 s

From TI flash to a fully surface-calibrated, metering device — over the air, no human touch.

±1.0%

All three self-calibrated meters vs. a 50-gallon marked-tank reference at 5 gpm.

Act I • Morning

The phantom in the pipeline

A bench meter “wedged” mid-test — stuck in an event, silently billing water with the valve closed. The autopsy caught it exactly: the reported flow rate was **bit-for-bit identical for 47 straight minutes** (real water is never bit-stable), fabricating **+200.14 gallons of phantom usage** while the raw sensor data underneath stayed perfectly alive.

Code archaeology traced it to a seven-line “glitch filter” shipped in TI firmware **v180, last Christmas Eve**: a substitution gate that, once the live measurement drifted 100 ns from its stored value, shipped the stale value forever — the condition causing the substitution was the condition preventing the cure. One cycle skip (500 ns, five times the gate) locked the trap permanently. Every fleet meter carries this code; the fleet has been protected for eight months only by an *accident* — the calibration machinery’s constant re-initializations kept curing the latch without anyone knowing it existed. It also retro-solved two standing mysteries: why the “self-heal” recovery step never once

worked, and why the recovery spiral “always returns 35” — the spiral’s wins were never about the value; they were accidental reboots of the trap.

Two-layer response, both shipped and live-fire tested the same day:

- **ST 17043 — defenses:** an impossible-water detector (bit-identical rate for 2 minutes → close the event, suppress the fossil), ingress rejection of degenerate sensor frames, a wall-clock event cap, a working remote force-close, and a pipeline watchdog. Validated live: the same disease that fabricated 200 gallons in the morning was contained to **0.13 gallons** in the afternoon reproduction.
- **TI v357 — the cure:** the substitution bounded and filter state invalidated on every config change. A deliberately hostile burst barrage that had frozen meters four times that day could no longer produce a single fossil. **Kill confirmed.**

Fleet follow-up queued: scan historical telemetry for the bit-identical fingerprint — several long-standing “constant-rate phantom” field cases now look suspiciously like this bug wearing a disguise.

Act II • Midday

Mapping the operating surface

The survey instrument — walk the meter across all 115 gain×envelope operating points and score each for cycle skips, errors, and noise — was promoted from a quarantine experiment to the production firmware line (**ST 17044**), with a hardened trigger, an RTC scheduler for unattended quiet-window surveys (`surveyAtHour / surveyEveryNDays`), and a TI-side hold (**v358**) so live calibration machinery no longer fights the walk.

Three maps taken across two days delivered the campaign’s central physics lesson: **the cliff moves**. The dead zone at low gain appeared, shrank, and moved between overnight, flowing, and evening conditions. A calibration that parks a meter based on one moment’s snapshot can park it on a cliff edge that only looks like solid ground in fair weather — which is precisely what legacy peak-seeking calibration does.

Side discoveries, all quantified for future work: radio check-ins blind the meter ~17% of the time at 10-minute cadence (samples discarded during sessions); “onset blindness” was caught directly in the data (the sensor briefly de-asserts its metering

flag at flow start); and the maximum legitimate flow slew was measured at ~28 ns/s — 18–20× below the smallest cycle skip, giving a clean physics threshold for telling water from glitches.

Act III · Evening

The walk is the cal

Then the design decision that reframed everything: don't use the map to *check* the calibration — **make the map the calibration**. No quick-cal, no peak-seeker, no host in the loop. TI v359: at every calibration, the meter walks its full surface locally (3 native-cadence samples per point, ~100 seconds total), finds the largest connected clean region, and commits to that region's deepest interior point — maximum margin from every measured cliff.

The first hardware run promptly delivered the perfect cautionary tale: a settle-timing bug poisoned the map, and the mathematically-correct picker parked the meter on the worst corner of the grid — the historical dead wall, temporarily disguised as clean by that evening's fair weather. Exactly the failure mode the operator's **trusted envelope** (gain 23–38 × env 30–50, the union of every cliff ever observed) had been drawn to prevent an hour earlier. Two rapid revisions closed it:

- **v360** — commits and recoveries constrained to the envelope; the full grid still walked for observability.
- **v361** — honest settling, a walk watchdog (never loop silently; commit best-available or fail loudly), and empty-map telemetry.

Recovery was redesigned in the same stroke: when error pressure hits, the meter no longer nudges blindly — it **flees to a random measured-clean cell inside the envelope**, stays where recovery succeeds, marks failed cells dirty (the map learns), and re-walks the surface when trusted territory runs out.

Map the territory → park with maximum margin → when glitched, jump somewhere known-good → when the map goes stale, re-map. The meter now owns its whole life cycle.

All three bench meters then calibrated themselves over the air — each choosing a *different* point, because each meter’s surface is genuinely individual:

METER	SELF-CHOSEN OPERATING POINT	IN ENVELOPE	50-GAL RUN VS. MARKED TANK
'8549	gain 29 / env 30	YES	49.48 gal (-1.0%)
'4423	gain 24 / env 50	YES	50.38 gal (+0.8%)
'3063	gain 33 / env 40	YES	50.18 gal (+0.4%)

Certification: 50.0 gallons of reference water at 5 gpm, all three meters within $\pm 1\%$, zero errors, zero recoveries, zero holds — on three machine-chosen operating points no human ever tuned.

Shipped today

RELEASE	SIDE	WHAT IT DOES	STATUS
ST 17043	STM32	Stuck-event defenses: impossible-water detector, ingress sentinel, event cap, watchdog, remote force-close, tunable radio cap	VALIDATED
ST 17044	STM32	Survey instrument on the fleet line + RTC survey scheduler	VALIDATED
TI v357	MSP430	Root-cause fix for the v180 substitution latch	KILL-CONFIRMED
TI v358	MSP430	Survey hold — calibration machinery idles during walks	VALIDATED
TI v359-360	MSP430	Surface cal + selection envelope (superseded same-day)	LESSONS
TI v361	MSP430	Surface cal, hardened — the certified build	CERTIFIED

Open items

- **Repeatability:** three mag-resets per meter — same region, same neighborhood, every time.
- **Fleet fossil scan:** quantify eight months of latch exposure in historical telemetry; re-examine the constant-rate phantom field cases.

- **Fleet roll plan:** v357+ must reach the fleet before any build that calms the recovery ladder (the ladder is currently the fleet's accidental protection).
- **Radio blackout buffer** (~17% blind time at dev cadence) and the register-save gap on resets — next ST revision.
- **Transient classifier:** the one surviving piece of the original re-acquisition agenda — “don't relocate for water” — thresholds already measured.
- **Longitudinal survey archive:** scheduled fleet surveys accumulate per-meter map history; the envelope eventually becomes each meter's own measured union-of-cliffs.

trio end state: ST 17044 + TI v361 · 10-min beats · event-end delivery

ledger: cal-diag-findings-0827.md